

A Beginner's Guide to Solar Energy

A clear introduction to how sunlight can be converted into electricity.

How solar panels work

Photovoltaic cells absorb light. The energy frees electrons inside the cell and creates direct-current electricity.

From panel to plug

1. Solar panels produce direct-current electricity.
2. An inverter changes it to alternating current used in buildings.
3. Electricity is used immediately, stored in a battery or supplied to the grid.

Benefits

Solar power is renewable during operation, can be installed at different scales and can reduce reliance on fossil fuels.

Points to consider

Output changes with daylight, weather, roof direction and shading. Manufacturing, transport and end-of-life recycling also require planning.

Connection to food

Solar electricity can support lighting, refrigeration, water pumping and other parts of a food system.

Learning check: Name the component that converts direct current into alternating current. Answer: the inverter.